

ALTA, Norway

WMO: 010490

Lat: 69.976N

Lon: 23.372E

Elev: 9

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-5.8	-1.8	-12.7	2.8	-5.3	-8.7	3.5	-1.1	29.7	31.0	25.2	30.1	7.1	210	0.746

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	72.7	59.8	68.7	58.4	65.2	56.8	61.4	69.8	59.3	67.1	57.5	64.4	9.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.3	70.0	65.8	55.4	65.3	63.5	53.7	61.3	61.1	27.4	70.1	26.0	67.2	24.8	64.5	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.4	20.5	17.8		-9.5	79.1	5.7	4.6	-13.6	82.4	-17.0	85.0	-20.2	87.6	-24.3	90.9
				-10.1	64.4	5.6	2.9	-14.1	66.5	-17.3	68.2	-20.4	69.8	-24.5	71.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.2	19.1	18.7	24.6	32.6	41.9	50.4	56.9	54.3	47.2	36.7	26.9	23.4	(6)
	DBStd	15.43	10.88	11.19	8.91	6.84	6.32	5.68	5.84	5.42	5.97	7.16	8.31	9.84	(7)
	HDD50	5541	958	876	787	522	263	60	5	15	123	414	692	826	(8)
	HDD65	10538	1423	1296	1252	972	716	438	263	334	534	877	1142	1291	(9)
	CDD50	493	0	0	0	0	12	73	218	149	39	2	0	0	(10)
	CDD65	14	0	0	0	0	0	1	10	3	0	0	0	0	(11)
	CDH74	87	0	0	0	0	0	5	66	16	0	0	0	0	(12)
	CDH80	14	0	0	0	0	0	0	13	1	0	0	0	0	(13)
(14) Wind	WSAvg	8.1	8.8	8.8	8.6	7.6	8.1	8.4	8.1	7.1	7.0	7.4	7.9	8.7	(14)
(15) Precipitation	PrecAvg	17.2	1.4	1.0	1.2	0.9	1.1	1.6	2.1	2.0	1.5	1.7	1.2	1.4	(15)
	PrecMax	23.2	2.8	2.0	2.0	1.9	2.4	3.5	6.1	3.9	2.9	3.3	2.9	3.5	(16)
	PrecMin	11.5	0.3	0.3	0.2	0.2	0.2	0.3	0.3	0.7	0.5	0.3	0.0	0.4	(17)
	PrecStd	2.7	0.7	0.4	0.5	0.4	0.6	0.8	1.3	0.8	0.6	0.8	0.7	0.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.2	43.4	45.0	52.1	65.8	73.0	81.2	76.1	65.9	56.2	46.4	46.3	(19)
		MCWB	38.1	38.0	39.0	43.1	52.6	57.5	65.0	61.6	56.8	49.6	41.6	40.7	(20)
	2%	DB	39.5	40.8	41.5	47.6	59.4	67.4	74.6	70.8	62.2	52.0	42.9	42.0	(21)
		MCWB	35.6	36.2	36.3	41.1	49.1	55.5	61.0	59.6	54.1	46.5	39.2	37.4	(22)
	5%	DB	36.2	37.2	38.9	44.5	55.0	63.0	70.3	66.7	59.0	48.6	40.2	38.8	(23)
		MCWB	32.9	33.6	34.3	39.1	46.4	53.7	59.4	58.0	52.8	44.0	36.7	34.8	(24)
	10%	DB	33.5	33.9	36.2	41.9	51.3	59.3	66.6	63.3	56.1	46.2	37.8	36.3	(25)
		MCWB	30.6	30.9	32.2	37.0	44.1	51.5	58.4	56.5	50.7	42.5	34.7	32.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.9	38.9	39.4	44.1	53.7	59.8	65.9	63.7	57.6	50.1	42.1	40.9	(27)
		MCDB	42.7	42.8	44.5	50.6	63.8	67.8	77.8	72.0	64.3	55.8	45.5	45.7	(28)
	2%	WB	35.7	36.3	36.8	41.4	49.8	56.8	62.4	60.8	55.1	46.9	39.2	37.4	(29)
		MCDB	39.2	40.3	41.2	46.7	58.7	66.0	72.1	68.5	60.4	51.5	42.7	41.4	(30)
	5%	WB	33.1	33.7	34.5	39.3	47.0	54.5	60.3	58.7	53.1	44.4	36.9	35.1	(31)
		MCDB	35.9	36.8	38.3	43.9	54.0	62.1	68.6	65.8	58.1	47.9	39.8	38.3	(32)
	10%	WB	30.5	31.0	32.5	37.4	44.7	52.3	58.6	56.8	51.1	42.6	34.8	33.0	(33)
		MCDB	33.2	33.7	36.0	41.5	50.4	58.4	65.9	62.8	55.6	45.9	37.6	36.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	10.9	11.8	11.0	10.4	9.8	10.7	10.9	10.0	8.3	9.2	10.2	(35)
		MCDBR	11.8	11.8	10.6	12.2	16.6	16.1	17.9	17.6	13.4	9.9	9.6	11.0	(36)
	5% WB	MCWBR	9.9	9.5	8.2	8.1	9.4	8.5	8.4	8.8	7.9	6.9	8.0	8.9	(37)
		MCWBR	11.7	11.7	10.5	11.4	15.7	14.8	16.0	15.5	12.1	9.4	9.2	11.1	(38)
(40) Clear-Sky Solar Irradiance	taub		N/A	0.248	0.300	0.323	0.339	0.341	0.349	0.334	0.306	0.258	N/A	N/A	(40)
		taud	N/A	2.044	2.115	2.178	2.303	2.464	2.478	2.503	2.533	2.191	N/A	N/A	(41)
	Ebn at Noon		0	146	218	254	266	271	264	254	230	143	0	0	(42)
		Edh at Noon	0	13	25	32	33	29	28	24	17	11	0	0	(43)
(44) All-Sky Solar Radiation	RadAvg		0	96	476	1077	1510	1540	1409	974	508	148	0	0	(44)
	RadStd		0	11	36	75	128	137	142	83	49	16	0	0	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page